Topic: Memory Segmentation in 8086

Q. Explain the concept of Memory Segmentation in the 8086 microprocessor. How is a 20-bit physical address calculated using a 16-bit segment register and a 16-bit offset?

> 📌 **Definition to Remember**
> **Memory Segmentation** divides the 1 MB of physical memory of the 8086 into logical **64 KB blocks (segments)**. Because internal registers are only 16-bit (max 64 KB), the CPU mathematically combines a 16-bit **Segment Base** and a 16-bit **Offset** to generate the required 20-bit physical address.

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1. The Four Segments

The CPU manages these memory blocks using four specialized 16-bit Segment Registers:
1. **Code Segment (CS):** Points to the executing program instructions. Paired with Instruction Pointer (IP).
2. **Data Segment (DS):** Points to program variables. Paired with BX, SI, or direct addresses.
3. **Stack Segment (SS):** Points to the system stack. Paired with Stack Pointer (SP) or Base Pointer (BP).
4. **Extra Segment (ES):** Additional data segment, often used as destination in string operations (paired with DI).

2. Logical vs Physical Address

- **Logical Address:** Format used by programmers `(Segment : Offset)`. e.g., `1000H:0002H`.
- **Physical Address:** The actual 20-bit address sent over the bus to access memory.

3. Calculating the 20-bit Physical Address

The Bus Interface Unit (BIU) performs this calculation:
Formula: `Physical Address = (Segment Register × 10H) + Offset`
(Multiplying by 10H in Hex shifts the segment value 4 bits to the left, making it a 20-bit base address).

Calculation Example:
* `CS` (Segment) = `2000H`
* `IP` (Offset) = `1234H`

1. **Shift Segment by 10H:** `2000H × 10H` = `20000H` (20-bit Base)
2. **Add Offset:**
   `20000H`
   `+ 1234H`
   `———————`
   `21234H` (20-bit Physical Address)

4. Advantages of Segmentation

- **Code Relocation:** Programs can be moved anywhere in memory by changing only the Segment Register; internal offset addresses remain unchanged.
- **Memory Protection/Separation:** Code, data, and stack are kept in separate segments, preventing accidental overwriting (e.g., data overwriting executable code).

⭐ Must-Write Points (for 10 marks)

1. 8086 has a 20-bit address bus (1 MB memory) but only 16-bit internal registers.
2. **Segmentation** divides memory into 64 KB logical blocks to solve this addressing mismatch.
3. Four Segment Registers: **CS (Code), DS (Data), SS (Stack), ES (Extra)**.
4. Physical Address is calculated by the BIU combining a Segment Base and an Offset.
5. **Formula**: `Physical Addr = (Segment Value × 10H) + Offset`.
6. Provide the calculation example (e.g., CS=2000H, IP=1234H → Base=20000H → Result=21234H).
7. **Advantages**: Easy program relocation and separation of code/data/stack.

⚡ Quick Recall

`Segmentation → 1 MB Memory / 16-bit registers → Divides into 64KB blocks (CS, DS, SS, ES) → Physical Addr = (Segment × 10H) + Offset → Pros: Relocation, Code/Data separation`